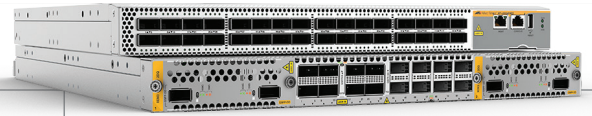


x980 Series

40/100/400 Gigabit Layer 3 Data Center Switches



Allied Telesis x980 Series powers data-driven environments, and support the demand for instant access to business information, always connected apps, and cloud-first networking, all in a scalable future-proof solution.

High-performing models come with 32 x 100/400G or 32 x 40/100G ports, and a rich data center and enterprise feature set that ensures flexible deployment.

Enable high-speed spine and leaf Clos network topology in the data center at up to 400Gbps – while the capacity and performance of the x980 Series also supports powerful enterprise core environments.

Data center performance

Ensure lossless low latency data forwarding from Top-of-Rack server deployment, with data center bridging features like Enhanced Transmission Selection (ETS) and Priority-based Flow Control (PFC).

The x980 Series provides always-on data availability and network resiliency with Multi-Chassis LAG (MLAG)¹, while workload mobility, virtual scalability, and seamless connectivity are provided by Ethernet VPN (EVPN)¹ and VXLAN².

Hyper-Converged Infrastructure (HCI)

The x980 Series has the performance and resilience to support hyperconverged infrastructure, where servers, storage, and switches connect seamlessly for data-driven environments that are software controlled.

This lowers cost and complexity while enabling better scalability and agility in managing and deploying IT services.

Network orchestration

The x980 Series comes pre-installed with the AlliedWare Plus™ operating system. They can also be deployed running the Community SONiC operating system – see your Allied Telesis representative for further information.

For easy integration into complex networks such as HCI that comprise physical, virtual, and multi-vendor devices, the x980 Series features:

- NETCONF/RESTCONF¹ APIs with OpenConfig YANG models.
- Ansible¹ Playbook libraries for automating common Day-1/Day-2 operations.

Network automation

Allied Telesis Autonomous Management Framework™ Plus (AMF Plus) meets the increasing management requirements of modern converged networks, by automating many everyday tasks. AMF Plus has powerful features that allow an entire network to be easily managed as a single virtual device.

Vista Manager™ is an intuitive graphical tool for monitoring and managing AMF Plus wired, Autonomous Wave Control (AWC) wireless, and third-party endpoint devices. Full visibility and powerful features enable proactive management.

Enterprise resiliency

Allied Telesis Virtual Chassis Stacking (VStack™), in conjunction with link aggregation, provides a network with no single point of failure and a resilient solution for high-availability applications.

For enterprise network core deployment, the x980 series can form a VStack of multiple units at any port speed, for enhanced resiliency and simplified device management. Stacks can also be created over long distance fiber links, making it the perfect choice for distributed environments too.

Allied Telesis Ethernet Protection Switched Ring (EPSRing™), and the standards-based G.8032 Ethernet Ring Protection, ensure that distributed network segments have high-speed, resilient access to online resources and applications.

Key Features



- 32 x 100/400G ports (x980-32DQ)
- 32 x 40/100G ports (x980-32CQ)
- AlliedWare Plus and SONiC modes of operation
- Data Center Bridging Capabilities Exchange Protocol (DCBX)
- Enhanced Transmission Selection (ETS)
- Priority-based Flow Control (PFC)
- Multi-Chassis LAG (MLAG)¹
- Ethernet VPN (EVPN)¹
- Virtual Extensible LAN (VXLAN)²
- Autonomous Management Framework Plus™ (AMF Plus)
- Virtual Chassis Stacking (VStack™)
- EPSRing™ and G.8032, ERPS for resilient rings
- Active Fiber Monitoring for fiber data and stacking links
- Link Monitoring
- NETCONF/RESTCONF¹ with OpenConfig YANG models
- Virtual Routing and Forwarding (VRF) Lite
- Bidirectional Forwarding Detection (BFD)
- Upstream Forwarding Only (UFO)

Reliable

Hot-swappable components such as fans, and load-sharing Power Supply Units (PSUs) pair with near hitless online stack reconfiguration, to ensure that maintenance doesn't affect network uptime.

¹ Available in a future software release

² VXLAN static tunnels supported. Dynamic tunnels available in a future software release

KEY FEATURES

Data center features

Data Center Bridging Capabilities Exchange Protocol (DCBX) lets network devices in the data center automatically discover and negotiate settings for DCB features, ensuring optimized performance.

Enhanced Transmission Selection (ETS) provides quality of service by allocating bandwidth to important traffic classes, with the flexibility to share remaining bandwidth to maximize traffic throughput and performance.

Priority-based Flow Control (PFC) enables lossless data forwarding by managing high priority traffic queues to support their class-of-service.

Multi-Chassis LAG (MLAG)¹ ensures resilience in data center networks by enabling multiple switches to act as a single logical device, coordinating their forwarding states to ensure load-balancing and redundancy.

Ethernet VPN (EVPN)¹ extends L2 connectivity across L3 networks providing a unified control plane, and partners with VXLAN as the data plane.

Virtual Extensible LAN (VXLAN)² tunnels join two or more L2 networks over an L3 IP network to form a single L2 broadcast domain. EVPN can automatically create VXLAN tunnels for flexibility and scalability in dynamic data center environments.

Autonomous Management Framework™ Plus (AMF Plus)

AMF Plus is a sophisticated suite of management tools that provide a simplified approach to network management. Common tasks are automated or made so simple that the everyday running of a network can be achieved without the need for highly-trained, and expensive, network engineers. Powerful features like centralized management, auto-backup, auto-upgrade, auto-provisioning and auto-recovery enable plug-and-play networking and zero-touch management.

The x980 Series operating as the AMF Plus master stores firmware and configuration backups for all other network nodes. This enables auto-provisioning and auto-upgrade by providing appropriate files to new network members.

An AMF Plus license provides all standard AMF network management and automation features, and also enables the AMF Plus intent-based networking features in Vista Manager EX.

Virtual Routing and Forwarding (VRF-Lite)

VRF-Lite provides Layer 3 network virtualization by dividing a single switch into multiple independent virtual routing domains. With independent routing domains, IP addresses can overlap without causing conflict, allowing multiple customers to have their own secure virtual network within the same physical infrastructure. VRF-Lite supports IPv4 and IPv6 unicast and multicast traffic.

The built-in DHCP Server on the x980 Series is VRF aware, enabling the supply of IP addresses to clients across multiple isolated networks

VCStack

Create a VCStack of multiple units at any port speed. Stacking links are connected in a ring so each device

has dual connections to further improve resiliency. VCStack provides a highly available system where network resources are spread out across stacked units, reducing the impact if one of the units fails. Aggregating switch ports on different units across the stack provides excellent network resiliency.

Long Distance Stacking

Long Distance stacking allows a VCStack to be created over fiber links to span longer distances, perfect for a distributed network environment.

EPSRing

EPSRing allows several switches to form protected rings with 50ms failover—perfect for high performance at the core of Enterprise or Provider Access networks. The x980 series can act as the EPSR Master.

SuperLoop Protection enables a link between two EPSR nodes to be in separate EPSR domains, improving redundancy and network fault resiliency.

G.8032 Ethernet Ring Protection

G.8032 provides standards-based high-speed ring protection, that can be deployed stand-alone, or interoperate with Allied Telesis EPSR.

Ethernet Connectivity Fault Monitoring (CFM) proactively monitors links and VLANs, and provides alerts when a fault is detected.

sFlow

sFlow is an industry-standard technology for monitoring high-speed switched networks. It provides complete visibility into network use, enabling performance optimization, usage accounting/billing, and defense against security threats. Sampled packets sent to a collector (up to 5 collectors can be configured) ensure it always has a real-time view of network traffic.

Active Fiber Monitoring

Active Fiber Monitoring prevents eavesdropping on fiber communications by monitoring received optical power. If an intrusion is detected, the link can be automatically shut down, or an operator alert can be sent. Active Fiber Monitoring is supported on fiber data and fiber stacking links

TACACS+ Command Authorization

TACACS+ Command Authorization offers centralized control over which commands may be issued by each specific AlliedWare Plus device user. It complements authentication and accounting services for a complete AAA solution.

VLAN Translation

Service Providers can use VLAN Translation to provide customer traffic with a unique VLAN-ID for use within the SP's network. It does this by mapping traffic arriving on a VLAN to a different VLAN on the outgoing paired interface.

This feature is also useful in Enterprise environments where it can be used to merge two networks together, without manually reconfiguring the VLAN numbering scheme.

Bi-Directional Forwarding Detection (BFD)

BFD enables fast detection of link failures, to minimize recovery. It works with static routes, and alongside BGP and OSPF routing protocols supporting faster shutdown of neighbor connections if a peer session goes down. When using VRF-Lite, BFD is supported globally or within a domain.

Upstream Forwarding Only (UFO)

UFO manages which ports in a VLAN can communicate with each other, and which only have upstream access to services, for secure multi-user deployment.

Multicast Source Discovery Protocol (MSDP)

MSDP enables two or more PIM-SM (Sparse Mode) domains to share information on active multicast sources, for more efficient forwarding of multicast traffic.

Link Monitoring (Linkmon)

Linkmon enables network health monitoring by regularly sending probes over key links to gather metrics comprising latency, jitter, and probe loss.

This supports pro-active network management and can also be used with triggers to automate a change to device or network configuration in response to the declining health of a monitored link.

NETCONF/RESTCONF

NETCONF/RESTCONF¹ with YANG data modeling provides a standardized way to represent data and securely configure devices, supporting existing management systems.

Software Defined Networking (SDN)

OpenFlow is a key technology for SDN orchestration. SDN controllers and other tools support automated behavior in a network and allow customized applications and services to be run.

Wireless network management

Integrated into the Device GUI, the wireless manager provides full network visibility of AMF Plus wired and AWC wireless devices. Manage and simplify wireless deployment, and provide a superior user experience with AWC automated wireless network optimization.

AMF Security and Application Proxy

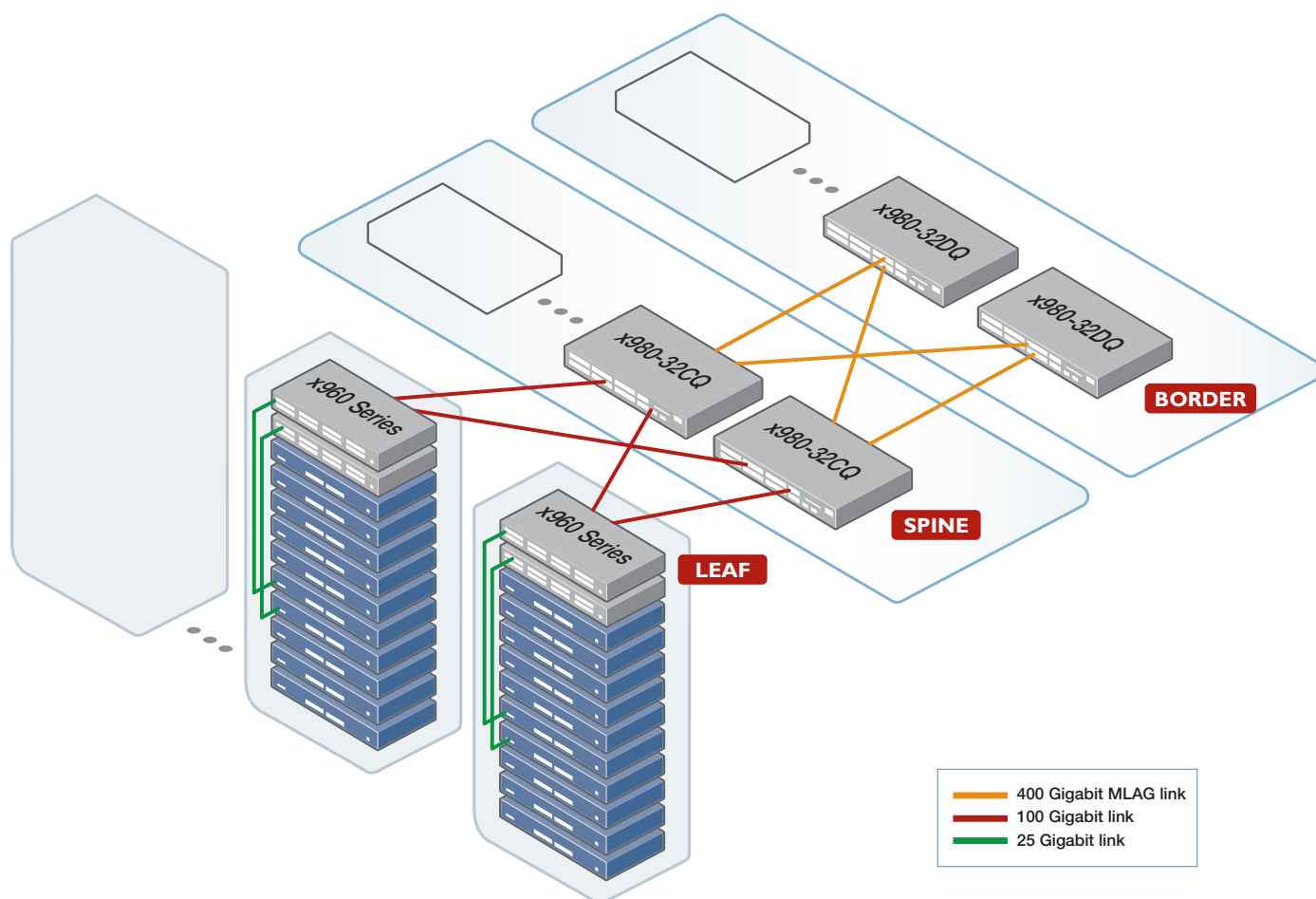
The AMF-Security (AMF-Sec) solution enables internal LAN threat detection and automatic end-point isolation to protect the network. The AMF Application Proxy (included for free in the base license) enables the AMF-Sec controller to communicate with the AMF Plus master when a threat is detected, so the AMF Plus master can take action to block the threat at source by quarantining the infected end-point.

¹ Available in a future software release

² VXLAN static tunnels supported. Dynamic tunnels available in a future software release

KEY SOLUTIONS

Data center spine and leaf connectivity



Today's businesses consume more data than ever, which has become critical to effective operation with always-on access to online resources and applications required. The move to using tools like generative AI and real-time collaboration has lifted the demand for increased performance and information delivery.

Data center Hyper-Converged Infrastructure (HCI) enables servers, storage, and switches to connect seamlessly in supporting data-driven environments. Centralized SDN management powers hardware and software resources which can be pooled as required. This lowers cost and complexity while enabling better scalability

and agility in managing and deploying IT services.

The x980 series enables high-performing spine and leaf connectivity at up to 400G. Partnered with x960 Series (available soon) Top of Rack (TOR) switches to provide 25G to servers, data transfer and information delivery is immediate.

Data Center Bridging Capabilities Exchange Protocol (DCBX) lets network devices in the data center automatically discover and negotiate settings for DCB features, ensuring optimized performance.

Ensure consistent bandwidth control and lossless forwarding with Enhanced

Transmissions Selection (ETS) and Priority-based Flow Control (PFC).

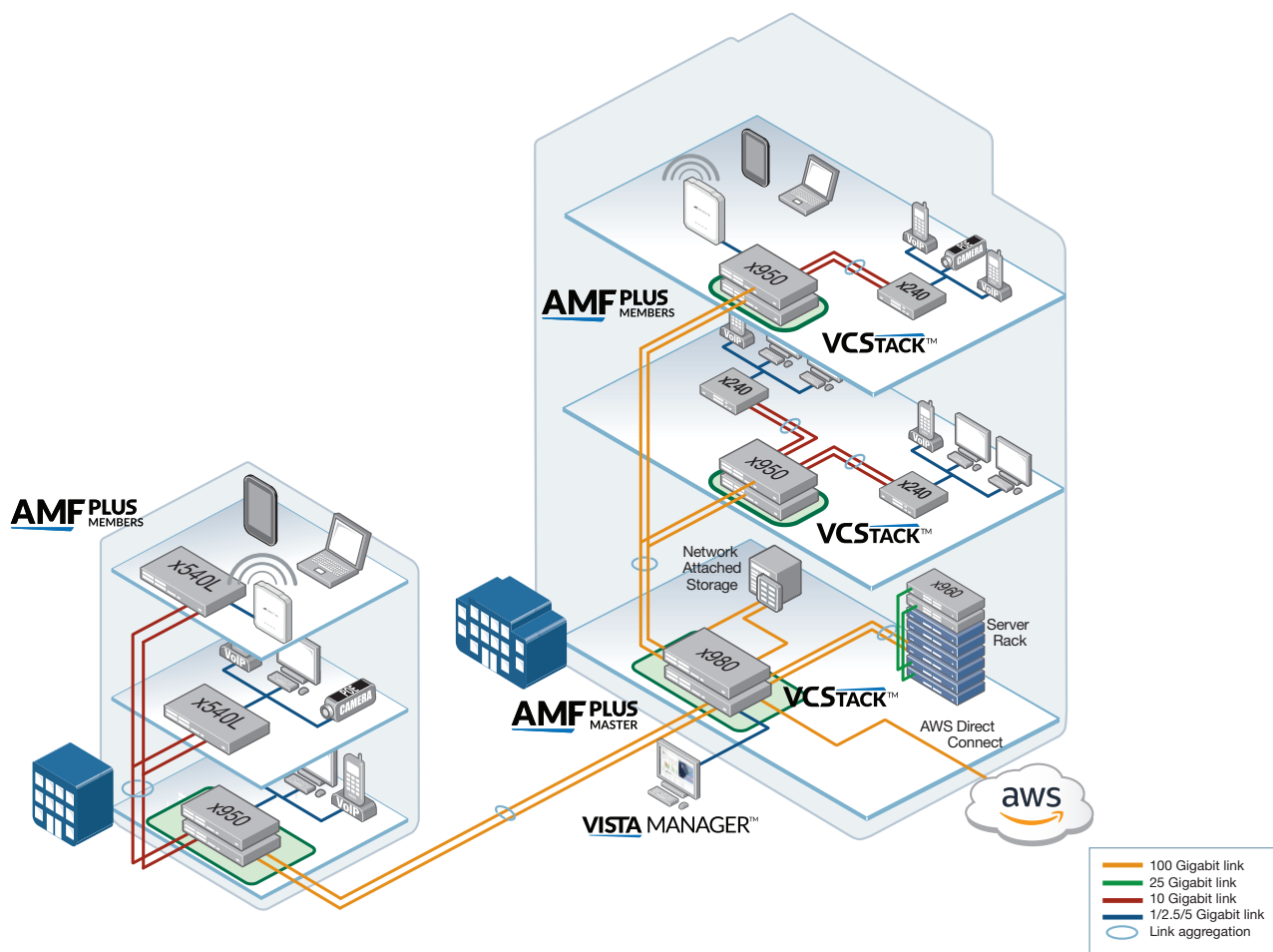
Network resiliency is assured using Multi-Chassis LAG (MLAG)¹, while workload mobility, virtual scalability, and seamless connectivity are provided by Ethernet VPN (EVPN)¹ and VXLAN².

As well as Allied Telesis innovative management tools like AMF Plus and Vista Manager, the x980 Series supports industry standard control protocols such as Netconf/Restconf¹, OpenFlow, and Ansible¹, so custom data center management applications and tools can be easily integrated.

¹ Available in a future software release

² VXLAN static tunnels supported. Dynamic tunnels available in a future software release

Enterprise network core



Today's enterprises demand ready access to online resources and applications, and require a high-performing network that can seamlessly carry multiple converged services. This 100G network core features the x980-32CQ (32 x 100G port) switch, and Virtual Chassis Stacking (VCSStack) for high availability, increased capacity and ease of management.

Using VCSStack at the core of the network allows multiple switches to appear as a single virtual chassis, simplifying management. In normal operation, the full

bandwidth of the network is used, ensuring always-available online services.

Using Amazon Web Services (AWS) Direct Connect ensures immediate access to cloud-based services, alongside locally stored business data and applications, to fully support modern hybrid-cloud environments.

AMF Plus allows the entire network to be unified for ease of management. The x980-32CQ acts as the AMF Plus Master, automatically backing up the entire

network, and enabling plug-and-play networking with zero-touch expansion and recovery

Vista Manager enables visual management and monitoring of the entire network including all wired, wireless, and endpoint devices.

The x980-32CQ delivers a protocol-less and Active/Active campus backbone solution, with high performance and flexible scalability.

SPECIFICATIONS

	10G SFP ports	40/100G QSFP28 ports	100/400 QSFP56-DD ports	Switching Fabric	Forwarding Rate
x980-32DQ	2	-	32	25.6Tbps	5.6Bpps
x980-32CQ	-	32	-	6.4Tbps	2.8Bpps

Physical Specifications

	Width	Depth	Height	Mounting	Weight
x980-32DQ	440 mm (17.32 in)	572 mm (22.51 in)	44 mm (1.73 in)	Rack-mount 1 RU	12.2 kg (26.9 lb)
x980-32CQ	440 mm (17.32 in)	490 mm (19.29 in)	44 mm (1.73 in)	Rack-mount 1 RU	9.4 kg (20.7 lb)

Power and Noise Characteristics

	Max Power Consumption	Max Heat Dissipation	Noise
x980-32DQ	994W	3,823 BTU/hr	78 dBA
x980-32CQ	447W	2,215 BTU/hr	66 dBA

Performance

- Cut-through switching for low-latency high-performance throughput
- Up to 160K MAC addresses
- Up to 1,000 static routes
- Up to 60,000 dynamic routes
- Up to 10KB jumbo frames
- 4K configurable VLANs
- 16GB DDR SDRAM
- 64GB SSD storage (x980-32DQ)
- 128GB SSD storage (x980-32CQ)

Reliability

- Modular AlliedWare Plus operating system
- Dual hot swappable PSUs with 1 + 1 redundancy
- Dual feed support: a separate power circuit can feed each power supply providing extra reliability
- Hot-swappable fan modules
- Full environmental monitoring of PSUs, fans, temperature and internal voltages, with SNMP traps to alert network managers in case of any failure

Diagnostic Tools

- Active Fiber Monitoring detects tampering on optical links
- Find-me device locator
- Connectivity Fault Management (CFM) for use with G.8032 ERPS
- Link Monitoring
- Automatic link flap detection and port shutdown
- Optical Digital Diagnostic Monitoring (DDM)
- Ping polling for IPv4 and IPv6

- Port mirroring
- No limit on mirrored ports
- Up to 7 mirror (analyzer) ports
- VLAN mirroring (RSPAN)
- TraceRoute for IPv4 and IPv6
- Uni-Directional Link Detection (UDLD)

IPv4 Features

- Black hole routing
- DHCP client, relay and server for IPv4
- Directed broadcast forwarding
- DNS relay
- Equal Cost Multi Path (ECMP) routing
- Policy-based routing
- Route maps
- Route redistribution (OSPF, BGP, RIP)
- Static IPv4 multicast routing
- Static unicast routing for IPv4
- UDP broadcast helper (IP helper)
- Virtual Routing and Forwarding Lite (VRF-Lite) up to 64 domains

IPv6 Features

- DHCPv6 client and relay
- DNS client and relay
- IPv4 and IPv6 dual stack
- IPv6 aware storm protection and QoS
- IPv6 hardware ACLs
- Device management over IPv6 networks with SNMP, Telnet and SSH
- IPv6 QoS support
- NTP client and server
- Static and dynamic IPv6 multicast routing
- Static and dynamic unicast routing for IPv6

- Log to IPv6 hosts with Syslog
- VRF-Lite

Management

- Autonomous Management Framework Plus (AMF plus) enables powerful centralized management and zero-touch device installation and recovery
- NETCONF/RESTCONF¹ northbound interface with YANG data modelling
- OpenAPI compliant RESTful APIs
- OpenFlow northbound interface
- Console management port on the front panel for ease of access
- Eco-friendly mode allows ports and LEDs to be disabled to save power
- Industry-standard CLI with context-sensitive help
- Powerful CLI scripting engine
- Comprehensive SNMP MIB support for standards-based device management
- Built-in text editor
- Event-based triggers allow user-defined scripts to be executed upon selected system events
- USB interface allows software release files, configurations and other files to be stored for backup and distribution to other devices
- Vista Mini for easy network-wide management
- Web-based Graphical User Interface (GUI)

Quality of Service (QoS)

- 8 priority queues with a hierarchy of high priority queues for real time traffic, and mixed scheduling, for each switch port
- Limit bandwidth per port or per traffic class down to 64kbps
- Wirespeed traffic classification with low latency essential for VoIP and real-time streaming media applications

¹ Available in a future software release

- Policy-based QoS based on VLAN, port, MAC and general packet classifiers
- Policy-based storm protection
- Extensive remarking capabilities
- Taildrop for queue congestion control
- Queue scheduling options for strict priority, weighted round robin or mixed scheduling
- IP precedence and DiffServ marking based on layer 2, 3 and 4 headers
- Data Center Bridging Capabilities Exchange Protocol (DCBX)
- Enhanced Transmission Selection (ETS)
- Priority-based Flow Control (PFC)

Resiliency Features

- Control Plane Prioritization (CPP) ensures the CPU always has sufficient bandwidth to process network control traffic
- Dynamic link failover (host attach)
- EPSR (Ethernet Protection Switched Rings)
- EPSRing SuperLoop Protection (SLP)
- Ethernet Ring Protection Switching (ITU-T G.8032 ERPS)
- Link aggregation (LACP) on LAN ports
- Loop protection: loop detection and thrash limiting
- Multi-Chassis LAG (MLAG)¹
- PVST+ compatibility mode
- Spanning Tree Protocols (STP, RSTP, MSTP)
- STP root guard
- VCSStack fast failover minimizes network disruption
- Flexi-stacking - use any port-speed to stack
- Long-distance stacking (LD-VCSStack)
- Virtual Router Redundancy Protocol (VRRP)

Security Features

- Access Control Lists (ACLs) based on layer 3 and 4 headers
- Configurable ACLs for management traffic
- Dynamic ACLs assigned via port authentication
- ACL Groups enable multiple hosts/ports to be included in a single ACL, reducing configuration
- Auth fail and guest VLANs
- RADIUS and TACACS+ Authentication, Authorisation and Accounting (AAA)
- Bootloader can be password protected for device security
- BPDU protection
- DHCP snooping, IP source guard and Dynamic ARP Inspection (DAI)
- DoS attack blocking and virus throttling
- Dynamic VLAN assignment
- Local RADIUS server for user and device authentication
- MAC-based authentication
- MAC address filtering and MAC address lock-down
- Network Access and Control (NAC) features manage endpoint security
- Port-based learn limits (intrusion detection)
- RADIUS group selection per VLAN or port
- RADIUS proxy
- Secure Copy (SCP)
- Secure File Transfer Protocol (SFTP) client
- Simple Certificate Enrollment Protocol (SCEP) supports secure management
- Strong password security and encryption
- Tri-authentication: MAC-based, web-based and IEEE 802.1x

VLAN Support

- Private VLANs provide security and port isolation for multiple customers using the same VLAN
- Upstream Forwarding Only (UFO)
- VLAN ID translation
- Voice VLAN

Environmental Specifications

- Operating temperature range:
0°C to 50°C (32°F to 122°F)
Derated by 1°C per 305 meters (1,000 ft)
- Storage temperature range:
-25°C to 70°C (-13°F to 158°F)
- Operating relative humidity range:
5% to 90% non-condensing
- Storage relative humidity range:
5% to 95% non-condensing
- Operating altitude:
3,000 meters maximum (9,843 ft)

Electrical approvals and compliances

- Electro Magnetic Compatibility: ICES-003 Issue 7 (2020), EN55032+EN55035, EN300386, BS55032/55035
- Electro Magnetic Interference: FCC Class A, EN55032 Class A, CISPR 32, VCCI Class A

Safety

- x980-32CQ: UL62368-1 ed.3, CB IEC/EN 62368-1 ed.2, TUV T-mark EN62368-1 2014
- x980-32DQ: UL62368-1 ed.3, CB IEC/EN 62368-1 ed.3, TUV T-mark EN IEC 62368-1 2020+A11

Restrictions on Hazardous Substances (RoHS) Compliance

- EU RoHS compliant
- China RoHS compliant-1

STANDARDS & PROTOCOLS

AlliedWare Plus Operating System

Version 5.5.6-1

Authentication

- RFC 1321 MD5 Message-Digest algorithm
- RFC 1828 IP authentication using keyed MD5

Border Gateway Protocol (BGP)

- BGP dynamic capability
- BGP outbound route filtering
- RFC 1772 Application of the Border Gateway Protocol (BGP) in the Internet
- RFC 1997 BGP communities attribute
- RFC 2385 Protection of BGP sessions via the TCP MD5 signature option
- RFC 2439 BGP route flap damping
- RFC 2545 Use of BGP-4 multiprotocol extensions for IPv6 inter-domain routing
- RFC 2858 Multiprotocol extensions for BGP-4
- RFC 2918 Route refresh capability for BGP-4
- RFC 3392 Capabilities advertisement with BGP-4
- RFC 3882 Configuring BGP to block Denial-of-Service (DoS) attacks
- RFC 4271 Border Gateway Protocol 4 (BGP-4)
- RFC 4360 BGP extended communities
- RFC 4456 BGP route reflection - an alternative to full mesh iBGP
- RFC 4724 BGP graceful restart
- RFC 4893 BGP support for four-octet AS number space
- RFC 5065 Autonomous system confederations for BGP
- RFC 5492 Capabilities advertisement with BGP-4
- RFC 5925 The TCP authentication option

¹ Available in a future software release

- RFC 6793 BGP support for four-octet AS number space
- RFC 7606 Revised error handling for BGP UPDATE messages

Cryptographic Algorithms

FIPS Approved Algorithms

Encryption (Block Ciphers):

- AES (ECB, CBC, CFB and OFB Modes)
- 3DES (ECB, CBC, CFB and OFB Modes)

Block Cipher Modes:

- CCM
- CMAC
- GCM
- XTS

Digital Signatures & Asymmetric Key Generation:

- DSA
- ECDSA
- RSA

Secure Hashing:

- SHA-1
- SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512)

Message Authentication:

- HMAC (SHA-1, SHA-2(224, 256, 384, 512)

Random Number Generation:

- DRBG (Hash, HMAC and Counter)

Non FIPS Approved Algorithms

- RNG (AES128/192/256)
- DES
- MD5

Ethernet Standards

- IEEE 802.2 Logical Link Control (LLC)
- IEEE 802.3 Ethernet
- IEEE 802.3ab1000BASE-T
- IEEE 802.3ae10 Gigabit Ethernet
- IEEE 802.3an10GBASE-T
- IEEE 802.3azEnergy Efficient Ethernet (EEE)
- IEEE 802.3ba 40GBASE-X
- IEEE 802.3bj 100GBASE-X
- IEEE 802.3bs 400GBASE-X
- IEEE 802.3by 25GBASE-X
- IEEE 802.3x Flow control - full-duplex operation
- IEEE 802.3z 1000BASE-X

IPv4 Features

- RFC 768 User Datagram Protocol (UDP)
- RFC 791 Internet Protocol (IP)
- RFC 792 Internet Control Message Protocol (ICMP)
- RFC 793 Transmission Control Protocol (TCP)
- RFC 826 Address Resolution Protocol (ARP)
- RFC 894 Standard for the transmission of IP datagrams over Ethernet networks
- RFC 919 Broadcasting Internet datagrams
- RFC 922 Broadcasting Internet datagrams in the presence of subnets
- RFC 932 Subnetwork addressing scheme
- RFC 950 Internet standard subnetting procedure
- RFC 951 Bootstrap Protocol (BootP)
- RFC 1027 Proxy ARP
- RFC 1035 DNS client
- RFC 1042 Standard for the transmission of IP datagrams over IEEE 802 networks
- RFC 1071 Computing the Internet checksum
- RFC 1122 Internet host requirements
- RFC 1191 Path MTU discovery
- RFC 1256 ICMP router discovery messages
- RFC 1518 An architecture for IP address allocation with CIDR
- RFC 1519 Classless Inter-Domain Routing (CIDR)
- RFC 1542 Clarifications and extensions for BootP
- RFC 1591 Domain Name System (DNS)
- RFC 1812 Requirements for IPv4 routers
- RFC 1918 IP addressing
- RFC 2581 TCP congestion control
- RFC 3021 Using 31-Bit Prefixes on IPv4 Point-to-Point Links

IPv6 Features

- RFC 1981 Path MTU discovery for IPv6
- RFC 2460 IPv6 specification
- RFC 2464 Transmission of IPv6 packets over Ethernet networks
- RFC 3484 Default address selection for IPv6

RFC 3587	IPv6 global unicast address format
RFC 3596	DNS extensions to support IPv6
RFC 4007	IPv6 scoped address architecture
RFC 4193	Unique local IPv6 unicast addresses
RFC 4213	Transition mechanisms for IPv6 hosts and routers
RFC 4291	IPv6 addressing architecture
RFC 4443	Internet Control Message Protocol (ICMPv6)
RFC 4861	Neighbor discovery for IPv6
RFC 4862	IPv6 Stateless Address Auto-Configuration (SLAAC)
RFC 5014	IPv6 socket API for source address selection
RFC 5095	Deprecation of type 0 routing headers in IPv6
RFC 5175	IPv6 Router Advertisement (RA) flags option
RFC 6105	IPv6 Router Advertisement (RA) guard

Management

AT Enterprise MIB including AMF Plus MIB and SNMP traps	
Optical DDM MIB	
SNMPv1, v2c and v3	
ANSI/TIA-1057 LLDP-Media Endpoint Detection	
IEEE 802.1AB Link Layer Discovery Protocol (LLDP)	
RFC 1155	Structure and identification of management information for TCP/IP-based Internets
RFC 1157	Simple Network Management Protocol (SNMP)
RFC 1212	Concise MIB definitions
RFC 1213	MIB for network management of TCP/IP-based Internets: MIB-II
RFC 1215	Convention for defining traps for use with the SNMP
RFC 1227	SNMP MUX protocol and MIB
RFC 1239	Standard MIB
RFC 1724	RIPv2 MIB extension
RFC 2578	Structure of Management Information v2 (SMIv2)
RFC 2579	Textual conventions for SMIv2
RFC 2580	Conformance statements for SMIv2
RFC 2674	Definitions of managed objects for bridges with traffic classes, multicast filtering and VLAN
RFC 2741	Agent extensibility (AgentX) protocol
RFC 2787	Definitions of managed objects for VRRP
RFC 2819	RMON MIB (groups 1,2,3 and 9)
RFC 2863	Interfaces group MIB
RFC 3176	sFlow: a method for monitoring traffic in switched and routed networks
RFC 3411	An architecture for describing SNMP management frameworks
RFC 3412	Message processing and dispatching for the SNMP
RFC 3413	SNMP applications
RFC 3414	User-based Security Model (USM) for SNMPv3
RFC 3415	View-based Access Control Model (VACM) for SNMP
RFC 3416	Version 2 of the protocol operations for the SNMP
RFC 3417	Transport mappings for the SNMP
RFC 3418	MIB for SNMP
RFC 3635	Definitions of managed objects for the Ethernet-like interface types
RFC 3636	IEEE 802.3 MAU MIB
RFC 4022	MIB for the Transmission Control Protocol (TCP)
RFC 4113	MIB for the User Datagram Protocol (UDP)
RFC 4188	Definitions of managed objects for bridges
RFC 4292	IP forwarding table MIB
RFC 4293	MIB for the Internet Protocol (IP)
RFC 4318	Definitions of managed objects for bridges with RSTP
RFC 4560	Definitions of managed objects for remote ping, traceroute and lookup operations
RFC 5424	The Syslog protocol
RFC 6020	YANG - A Data Modeling Language for the Network Configuration Protocol (NETCONF) ¹
RFC 6241	Network Configuration Protocol (NETCONF)
RFC 6244	Architecture for Network Management Using NETCONF and YANG
RFC 6527	Definitions of managed objects for VRRPv3
RFC 7950	The YANG 1.1 Data Modeling Language
RFC 8040	RESTCONF Protocol

Multicast Support

Bootstrap Router (BSR) mechanism for PIM-SM	
IGMP query solicitation	
IGMP snooping (IGMPv1, v2 and v3)	
IGMP snooping fast-leave	
IGMP/MLD multicast forwarding (IGMP/MLD proxy)	
MLD snooping (MLDv1 and v2)	
PIM and PIM SSM for IPv6	
RFC 1112	Host extensions for IP multicasting (IGMPv1)
RFC 2236	Internet Group Management Protocol v2 (IGMPv2)
RFC 2710	Multicast Listener Discovery (MLD) for IPv6
RFC 2715	Interoperability rules for multicast routing protocols
RFC 3306	Unicast-prefix-based IPv6 multicast addresses
RFC 3376	IGMPv3
RFC 3590	Source address selection for the Multicast Listener Discovery (MLD) protocol
RFC 3618	Multicast Source Discovery Protocol (MSDP)
RFC 3810	Multicast Listener Discovery v2 (MLDv2) for IPv6
RFC 3956	Embedding the Rendezvous Point (RP) address in an IPv6 multicast address
RFC 3973	PIM Dense Mode (DM)
RFC 4541	IGMP and MLD snooping switches
RFC 4604	Using IGMPv3 and MLDv2 for source-specific multicast
RFC 4607	Source-specific multicast for IP
RFC 7761	Protocol Independent Multicast - Sparse Mode (PIM-SM)

Open Shortest Path First (OSPF)

OSPF link-local signaling	
OSPF MD5 authentication	
Out-of-band LSDB resync	
RFC 1245	OSPF protocol analysis
RFC 1246	Experience with the OSPF protocol
RFC 1370	Applicability statement for OSPF
RFC 1765	OSPF database overflow
RFC 2328	OSPFv2
RFC 2370	OSPF opaque LSA option
RFC 2740	OSPFv3 for IPv6
RFC 3101	OSPF Not-So-Stubby Area (NSSA) option
RFC 3509	Alternative implementations of OSPF area border routers
RFC 3623	Graceful OSPF restart
RFC 3630	Traffic engineering extensions to OSPF
RFC 4552	Authentication/confidentiality for OSPFv3
RFC 5329	Traffic engineering extensions to OSPFv3
RFC 5340	OSPFv3 for IPv6 (partial support)

Quality of Service (QoS)

IEEE 802.1p Priority tagging	
IEEE 802.1Qaz Data Center Bridging Capabilities Exchange Protocol (DCBX)	
IEEE 802.1Qaz Enhanced Transmission Selection (ETS)	
IEEE 802.1Qbb Priority-based Flow Control (PFC)	
RFC 2211	Specification of the controlled-load network element service
RFC 2474	DiffServ precedence for eight queues/port
RFC 2475	DiffServ architecture
RFC 2597	DiffServ Assured Forwarding (AF)
RFC 2697	A single-rate three-color marker
RFC 2698	A two-rate three-color marker
RFC 3246	DiffServ Expedited Forwarding (EF)

Resiliency Features

ITU-T G.8032 / Y.1344 Ethernet Ring Protection Switching (ERPS)	
IEEE 802.1agConnectivity Fault Management (CFM), Continuity Check Protocol (CCP)	
IEEE 802.1AX Link aggregation (static and LACP)	
IEEE 802.1D MAC bridges	
IEEE 802.1s Multiple Spanning Tree Protocol (MSTP)	
IEEE 802.1w Rapid Spanning Tree Protocol (RSTP)	
IEEE 802.3ad Static and dynamic link aggregation	
RFC 5798	Virtual Router Redundancy Protocol version 3 (VRRPv3) for IPv4 and IPv6

Routing Information Protocol (RIP)

RFC 1058	Routing Information Protocol (RIP)
RFC 2080	RIPng for IPv6

RFC 2081	RIPng protocol applicability statement
RFC 2082	RIP-2 MD5 authentication
RFC 2453	RIPv2

Security Features

SSH remote login	
SSLv2 and SSLv3	
IEEE 802.1X authentication protocols (TLS, TTLS, PEAP and MD5)	
IEEE 802.1X multi-suplicant authentication	
IEEE 802.1X port-based network access control	
RFC 2560	X.509 Online Certificate Status Protocol (OCSP)
RFC 2818	HTTP over TLS ("HTTPS")
RFC 2865	RADIUS authentication
RFC 2866	RADIUS accounting
RFC 2868	RADIUS attributes for tunnel protocol support
RFC 2986	PKCS #10: certification request syntax specification v1.7
RFC 3546	Transport Layer Security (TLS) extensions
RFC 3579	RADIUS support for Extensible Authentication Protocol (EAP)
RFC 3580	IEEE 802.1x RADIUS usage guidelines
RFC 3748	PPP Extensible Authentication Protocol (EAP)
RFC 4251	Secure Shell (SSHv2) protocol architecture
RFC 4252	Secure Shell (SSHv2) authentication protocol
RFC 4253	Secure Shell (SSHv2) transport layer protocol
RFC 4254	Secure Shell (SSHv2) connection protocol
RFC 5176	RADIUS Change of Authorization (CoA)
RFC 5280	X.509 certificate and Certificate Revocation List (CRL) profile
RFC 5425	Transport Layer Security (TLS) transport mapping for Syslog
RFC 5656	Elliptic curve algorithm integration for SSH
RFC 6125	Domain-based application service identity within PKI using X.509 certificates with TLS
RFC 6614	Transport Layer Security (TLS) encryption for RADIUS
RFC 6668	SHA-2 data integrity verification for SSH
RFC 8446	Transport Layer Security (TLS) v1.3
RFC 8894	Simple Certificate Enrollment Protocol (SCEP)

Services

RFC 854	Telnet protocol specification
RFC 855	Telnet option specifications
RFC 857	Telnet echo option
RFC 858	Telnet suppress go ahead option
RFC 1091	Telnet terminal-type option
RFC 1350	Trivial File Transfer Protocol (TFTP)
RFC 1985	SMTP service extension
RFC 2049	MIME
RFC 2131	DHCPv4 (server, relay and client)
RFC 2132	DHCP options and BootP vendor extensions
RFC 2616	Hypertext Transfer Protocol - HTTP/1.1
RFC 2821	Simple Mail Transfer Protocol (SMTP)
RFC 2822	Internet message format
RFC 3046	DHCP relay agent information option (DHCP option 82)
RFC 3315	DHCPv6 (server, relay and client)
RFC 3396	Encoding long options in DHCPv4
RFC 3633	IPv6 prefix options for DHCPv6
RFC 3646	DNS configuration options for DHCPv6
RFC 3993	Subscriber-ID suboption for DHCP relay agent option
RFC 4330	Simple Network Time Protocol (SNTP) version 4
RFC 4954	SMTP service extension for authentication
RFC 5905	Network Time Protocol (NTP) version 4

VLAN Support

Generic VLAN Registration Protocol (GVRP)	
IEEE 802.1ad Provider bridges (VLAN stacking, Q-in-Q)	
IEEE 802.1Q Virtual LAN (VLAN) bridges	
IEEE 802.1v VLAN classification by protocol and port	
IEEE 802.3ac VLAN tagging	
Static VXLAN tunnels (part of RFC 7348)	

Voice over IP (VoIP)

LLDP-MED ANSI/	
Voice VLAN	

FEATURE LICENSES

	Description	Includes	Stack Licensing
AT-FL-x980-01	x980 Premium license	■ OSPF (60,000 routes)³ ■ BGP (60,000 routes)³ ■ PIMv4-SM, DM and SSM (2,000 entries) ■ RIPv4 (8,000 routes) ■ OSPFv3 (30,000 routes) ■ BGP4+ (30,000 routes) ■ PIMv6-SM and SSM (1,000 entries) ■ MLDv1 and v2 ■ RADIUS Full ■ UDLD ■ VLAN double tagging (Q-in-Q) ■ VLAN Translation ■ VXLAN ■ PTP Transparent Mode ■ VRF lite (64 domains)	■ One license per stack member
AT-SW-APM10-1YR^{4,5}	Cumulative AMF Plus Master license	■ AMF Plus Master license for up to 10 nodes for 1 year	■ One license per stack
AT-SW-APM10-5YR^{4,5}	Cumulative AMF Plus Master license	■ AMF Plus Master license for up to 10 nodes for 5 years	■ One license per stack
AT-SW-APC10-1YR^{4,6}	Cumulative AMF Plus Controller license	■ AMF Plus Controller license for up to 10 areas for 1 year	■ One license per stack
AT-SW-APC10-5YR^{4,6}	Cumulative AMF Plus Controller license	■ AMF Plus Controller license for up to 10 areas for 5 years	■ One license per stack
AT-FL-x980-OF13-1YR	OpenFlow license	■ OpenFlow v1.3 for 1 year	■ Not supported on a stack
AT-FL-x980-OF13-5YR	OpenFlow license	■ OpenFlow v1.3 for 5 years	■ Not supported on a stack
AT-FL-x980-VLF	VRF-Lite Full license	■ VRF lite (600 domains)	■ One license per stack member

³ 64 OSPF and BGP routes included in base license

⁴ AMF Plus licenses provide all standard AMF network management and automation features. They also enable AMF Plus intent-based networking features in Vista Manager EX

⁵ Purchase one license per 10 nodes (up to 180 nodes maximum)

⁶ Purchase one license per 10 areas (up to 60 areas maximum)

ORDERING INFORMATION

x980-32DQ

AT-x980-32DQ	Lossless, Low-latency Data center Layer 3 switch with 32 x 100G/400G QSFP28/QSFP-DD ports, dual hot-swappable PSUs, and 6 x Fan modules bays. (6 x front-to-back airflow fan modules included. 2 x PSUs need to be ordered separately).
AT-PWRSYS5-xy0⁷	1600W AC system power supply, Front-to-back airflow
AT-PWRSYS5-x80⁷	1600W DC system power supply, Front-to-back airflow

x980-32CQ

AT-x980-32CQ	Lossless, Low-latency Data center Layer 3 switch with 32 x 40G/100G QSFP+/QSFP28 ports, dual hot-swappable PSUs, and 5 x Fan modules bays. (5 x front-to-back airflow fan modules included. 2 x PSUs need to be ordered separately).
AT-PWRSYS4-xy0^{7,8}	1600W AC system power supply, Front-to-back airflow
AT-PWRSYS4-x80^{7,8}	1600W DC system power supply, Front-to-back airflow
AT-PWRSYS4R-xy0^{7,8}	1600W AC system power supply, Back-to-front reverse airflow
AT-PWRSYS4R-x80^{7,8}	1600W DC system power supply, Back-to-front reverse airflow
AT-FAN17	Hot-swappable fan module, Front-to-back airflow
AT-FAN17R⁸	Hot-swappable fan module, Back-to-front reverse airflow

⁷ Where x = 0 for non-TAA and 9 for TAA compliant

Where y = 1 for US power cord

3 for UK power cord

4 for AU power cord

5 for EU power cord

⁸ PSUs used in the switch must be of the identical part

number. Using non-matching PSUs voids the product warranty.

Accessories

400G QSFP56-DD Modules	
AT-QSFPDD-SR8 ⁹	400G, QSFPDD, MPO, 850 nm short-haul up to 100 m with MMF
AT-QSFPDD-1CU	1 meter QSFPDD direct attach cable
AT-QSFPDD-2.5CU	2.5 meter QSFPDD direct attach cable
100G QSFP28 Modules	
AT-QSFP28SR4	100GSR 850nm short-haul up to 100 m with MMF
AT-QSFP28LR4	100GLR 1310nm medium-haul, 10 km with SMF
AT-QSFP28ZR4	100GZR 1310 nm long-haul, 80 km with SMF, TAA ¹⁰
AT-QSFP28-1CU	1 meter QSFP28 direct attach cable
AT-QSFP28-3CU	3 meter QSFP28 direct attach cable
40G QSFP+ Modules	
AT-QSFPSR4	40GSR4 850 nm short-haul up to 150 m with MMF, MPO-12
AT-QSFPSR4LCa	40GSR4 850 nm short-haul up to 150 m with MMF, LC
AT-QSFPLR4	40GLR4 1310 nm medium-haul, 10 km with SMF

AT-QSFPER4	40GER4 1310 nm long-haul, 40 km with SMF
AT-QSFP1CU	1 meter QSFP+ direct attach cable
AT-QSFP3CU	3 meter QSFP+ direct attach cable
10GbE SFP+ Modules	
AT-SP10SR	10GSR 850 nm short-haul, 300 m with MMF
AT-SP10SR/I	10GSR 850 nm short-haul, 300 m with MMF industrial temperature
AT-SP10LRM	10GLRM 1310 nm short-haul, 220 m with MMF
AT-SP10LR	10GLR 1310 nm medium-haul, 10 km with SMF
AT-SP10LRa/I	10GLR 1310 nm medium-haul, 10 km with SMF industrial temperature
AT-SP10ER20/I	10GER 1310 nm long-haul, 20 km with SMF industrial temperature
AT-SP10ER40a/I	10GER 1310 nm long-haul, 40 km with SMF industrial temperature

⁹ Available soon

¹⁰ Trade Act Agreement compliant